VRAAG / *QUESTION* 1

'n Deeltjie beweeg in 'n reguitlyn in die x-as rigting en die grootte van sy snelheid is v.
A particle is moving in a straight line in the x-direction and the magnitude of its velocity is v.

(a) Definieer sy snelheid v en gee die eenheid. / *Define its velocity v and give its unit.*

____ (2)

(b) Met $v = v_0 + at$, toon deur middel van integrasie aan dat vir 'n verplasing tussen x_0 en x in 'n tyd t ($t_0 = 0$) is $x - x_0 = v_0 t + \frac{1}{2}at^2$.

With $v = v_0 + at$, *show by means of integration that for a displacement between* x_0 *and* x *in a time* t ($t_0 = 0$) *is* $x - x_0 = v_0 t + \frac{1}{2}at^2$.

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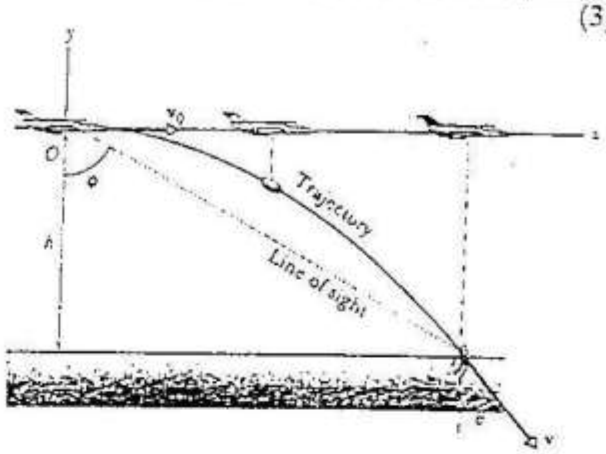
____ (3)

(c) 'n Vliegtuig vlieg teen v_0 = 430 km/h op 'n hoogte van 1200 m bo die see en laat 'n reddingspak val vir 'n drenkeling in die see.

An aeroplane has a speed v_0 = 430 km/h *and is flying at a height of* 1200 m *above the sea. It drops a parcel to a person in the sea.*



(i) Toon aan dat die tyd wat die reddingspak deur die lug trek is t = 15,65 s.
Show that the time of flight for the parcel is t = 15,65 s.

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____ (3)

Vraag / *Question* 1(c)(ii)(vervolg)(*continue*) / …

(ii) Toon aan dat die horisontale afstand wat die reddingspak afgelê het is 1869 m.
Show that the horizontal distance covered by the parcel is 1869 m.

(2)

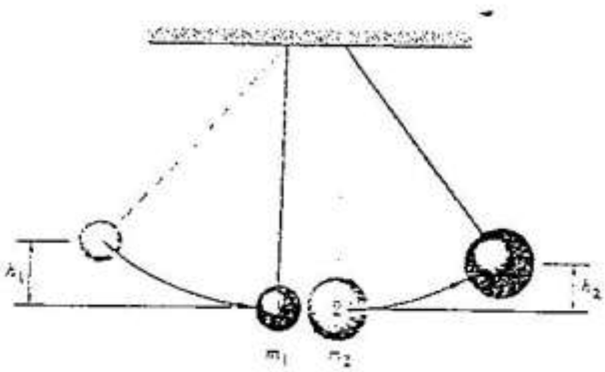
(iii) Bepaal die hoek ϕ. / *Determine the angle* ϕ.

(1)

VRAAG / *QUESTION* 2

Twee metaal balle is vas aan toue soos in meegaande skets. Bal 1 (massa m_1 = 30 gram) word opgelig tot hoogte h_1 = 8,0 cm en dan gelos om elasties te bots met bal 2 met massa m_2 = 75 gram.

Two metal spheres are suspended by cords as shown in the figure. Sphere 1 (mass m_1 = 30 *gram) is pulled to a height* h_1 = 8,0 cm *and then released. It undergoes an elastic collision with sphere 2 whose mass is* m_2 = 75 gram.



(a) Toon aan dat spoed die v_{1i} van 1 net voor die botsing is v_{1i} = 1,252 m/s.
Show that the speed v_{1i} *of* 1 *just before the collision is* v_{1i} = 1,252 m/s.

(3)

(b) Na die botsing is v_{1f} = - 0,537 m/s. Toon aan dat v_{2f} = 0,7156 m/s.
After the collision v_{1f} = - 0,537 m/s. *Show that* v_{2f} = 0,7156 m/s.

(c) Bepaal die hoogte h_2 van m_2 na die botsing.
To what height h_2 *does 2 swing after the collision?*

(3)

VRAAG / *QUESTION* 3

(a) Definieer hoeksnelheid ω en gee die eenhede daarvan.
Define angular velocity ω *and give its units.*

(2)

(b) (i) 'n Slypwiel begin draai (vanuit rus) met 'n konstante hoekversnelling $\alpha = 0{,}35\ \text{rad/s}^2$. Bepaal die hoekverplasing θ wat die wiel in 18 s maak.
A grindstone starts from rest and has a constant angular acceleration $\alpha = 0{,}35\ \text{rad/s}^2$. *What is the angular displacement* θ *at* $t = 18$ s?

(3)

(ii) Hoeveel omwentelinge het die wiel gemaak in 18 s?
Determine the number of revolutions completed in the 18 s.

(1)

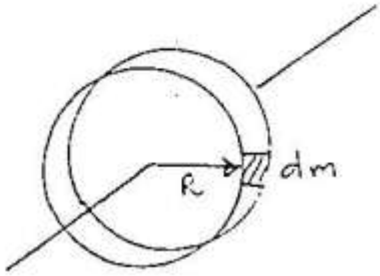
(iii) Wat is die wiel se hoeksnelheid na 18 s?
What is the wheel's angular velocity at $t = 18$ s?

(2)

Vraag / *Question* 3(c)(vervolg)(*continue*) ...

(c) 'n Dun ring met massa m en straal R roteer om 'n as deur sy middelpunt. Toon aan dat die traagheidsmoment van die hele ring is $I = mR^2$.

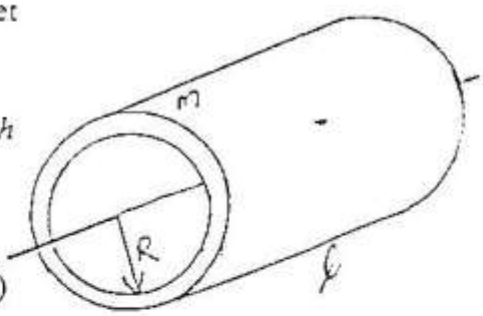
A thin ring has a mass m *and radius* R *and it rotates about an axis through its center. Show that the moment of inertia of the whole ring is* $I = mR^2$.



(2)

(d) Wat is die traagheidsmoment van 'n dun silinder met straal R, massa m en lenge ℓ?

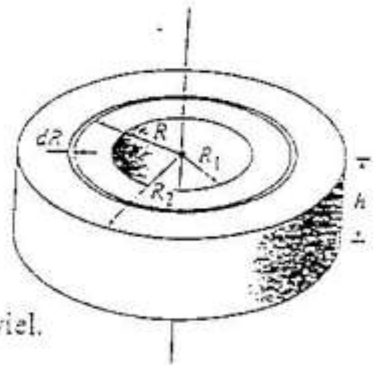
What is the moment of inertia of a thin cylinder with radius R, *mass* m *and length* ℓ?



(1)

(e) 'n Slypwiel het 'n gat in die middel met 'n binne radius R_1, 'n buiteradius R_2 en 'n massa M.

A grindstone with a hole in the center has an inner radius R_1, *an outer radius* R_2 *and a mass* M.



(i) Skryf 'n uitdrukking neer vir die volume V van die wiel.
Give an expression for the volume V *of the wheel.*

(1)

(ii) Beskou die wiel as opgebou uit dun ringe met dikte dR. Gee 'n uitdrukking vir die volume dV van 'n ring met straal R, dikte dR en hoogte h.

The cylinder can be constructed from a number of thin concentric cylindrical rings of thickness dR. *Give an expression for the volume* dV *of a ring with radius* R, *thickness* dR *and height* h.

(1)

(iii) 'n Dun ring met massa dm, volume dV en dikte dR se digtheid is ρ. Gee 'n uitdrukking vir dm in terme van ρ.

The mass of a thin ring is dm, *its volume is* dV *and its thickness is* dR. *The ring has a density* ρ. *Give an expression for* dm *in terms of* ρ.

(2)

(iv) Met bostaande uitdrukking vir dm, herlei 'n uitdrukking vir die traagheidsmoment I vir die wiel in terme van R_1 en R_2.

With the expression for dm *above, derive an expression for the moment of inertia* I *for the whole wheel in terms of* R_1 *and* R_2.

(5)

(v) Indien die totale massa van die wiel M is en sy digtheid is ρ, skryf 'n uitdrukking vir ρ in terme van M, R_1 en R_2.

The wheel has a total mass M *and density* ρ. *Write an expression for* ρ *in terms of* M, R_1 *and* R_2.

(1)

(vi) Met $R_2^4 - R_1^4 = (R_2^2 - R_1^2)(R_2^2 + R_1^2)$, toon aan dat $I = \frac{1}{2} M (R_2^2 + R_1^2)$.

With $R_2^4 - R_1^4 = (R_2^2 - R_1^2)(R_2^2 + R_1^2)$, *show that* $I = \frac{1}{2} M (R_2^2 + R_1^2)$.

(3)

(vii) Die wiel het 'n massa M = 5 kg met R_1 = 0,1 m en R_2 = 0,5 m. Bepaal die waarde van I.

If the wheel has a mass M = 5 kg, *with* R_1 = 0,1 m *and* R_2 = 0,5 m, *determine the value of* I.

(2)

(viii) Die wiel roteer met 'n hoeksnelheid ω = 1000 omwentelings per minuut. Bepaal die hoeksnelheid in rad/s.

The wheel is rotating with an angular velocity ω = 1000 *revolutions per minute. Determine the angular velocity in rad/s.*

(2)

(ix) Bepaal die kinetiese energie wat in die wiel gestoor is.

What is the kinetic energy stored in the wheel?

(2)

(x) Indien al die rotasie kinetiese energie oorgedra word aan 'n 60 W gloeilamp, hoe lank sal die gloeilamp brand?

If all the rotational kinetic energy is transferred to a 60 W globe, for how long will the lamp light up?

(3)

VRAAG / *QUESTION* 4

(a) 'n Veer word oor 'n afstand x saamgedruk. Gee Hooke se Wet vir die krag van die veer.
A spring is compressed over a distance x. *Give Hooke's law for the spring force.*

(1)

(b) Gee die formule om die arbeid W te bepaal wanneer 'n konstante krag $\overline{F}$ oor 'n afstand x verplaas.
Give a formula for the work W done when a force $\overline{F}$ *acts over a distance* x.

(1)

(c) Gee 'n formule vir die arbeid W wat verrig word deur 'n krag $\overline{F}(x)$, wat nie konstant is nie, tussen die punte x_i en x_f.
Give a formula for the work W *done by a force* $\overline{F}(x)$, *which is not constant, between two points* x_i *and* x_f.

(1)

(d) Toon aan dat die arbeid wat verrig word deur die krag $\overline{F}(x)$ van 'n veer gegee word deur $W = -\frac{1}{2}kx^2$ indien $x_i = 0$ en $x_f = x$.

Show that the work doen by a spring force $\overline{F}(x)$ *is given by* $W = -\frac{1}{2}kx^2$ *if* $x_i = 0$ *and* $x_f = x$.

(3)

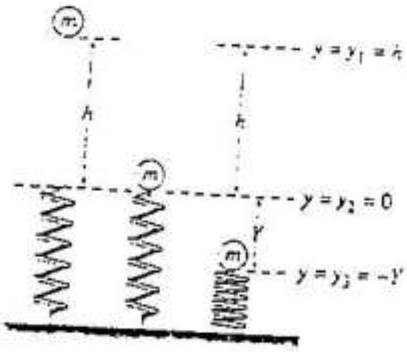
(e) Indien die verandering in potensiële energie $\Delta U = -W$, wat is die potensiële energie wat gestoor is in 'n veer wat 'n afstand x saamgeduk is?
If the change in potential energy $\Delta U = -W$, *what is the potential energy stored in a spring which is compressed a distance* x.

(1)

Vraag / *Question* 4(f)(vervolg / *continue*) / ...

(f) 'n Bal met massa m = 2,60 kg val vertikaal afwaarts vanuit rus deur 'n hoogte h = 55 cm. Dit val op 'n veer wat dan Y = 15 cm saamdruk.

A ball with mass m = 2,60 kg, *starting from rest, falls a vertical distance* h = 55 cm *before striking a coiled spring, which compresses an amount* Y = 15 cm.



(i) Toon met behoud van energie aan dat die snelheid v_2 van die bal net wanneer dit die veer tref (by posisie $y_2 = 0$) is 3,28 m/s.

From the conservation of energy show that the velocity v_2 *of the ball just as it touches the spring (at position* $y_2 = 0$*) is* 3,28 m/s.

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____ (4)

(ii) Bepaal die veerkonstante k indien die veer ten volle saamgedruk word tot by $y_3 = -Y = -0{,}15$ m

Determine the spring constant k *if the spring is fully compressed at* $y_3 = -Y = -0{,}15$ m

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____ (5)

Vraag / *Question* 4(g)(vervolg / *continue*) /

(g) Met veerkonstante $k = 1583$ N/m, bepaal die posisie(s) y_4 (relatief tot y_2) waar $v_4 = 3$ m/s. Indien u meer as een antwoord vir y_4 kry, verduidelik kortliks hul betekenis.

If the spring constant $k = 1583$ N/m, *determine the position(s)* y_4 *(relative to* y_2*) where* $v_4 = 3$ m/s. *If you find more than one answer for* y_4, *give a short description of their meaning.*

(6)

VRAAG/*QUESTION* 5

(a) Kragte wat op 'n liggaam inwerk is sodanig dat $\sum \overline{F} \neq 0$. Gee Newton se tweede wet.
For different forces acting on an object $\sum \overline{F} \neq 0$. *Give Newton's second law.*

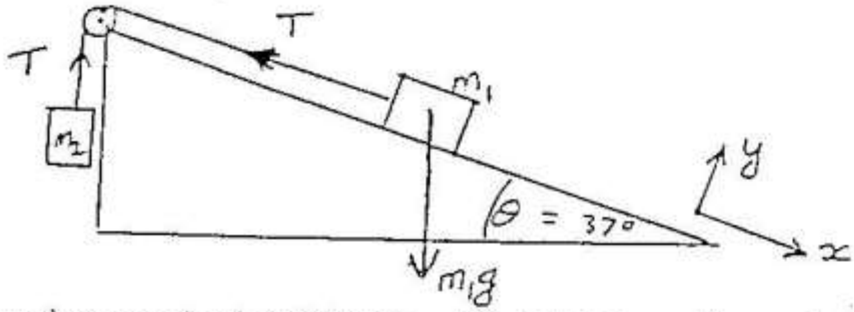
(1)

(b) Gee 'n formule vir die statiese wrywingskrag f_s indien die wrywingskoëffisiënt μ_s is.
Give a formula for the static frictional force f_s *with the static coefficient of friction* μ_s.

(2)

Vraag / *Question* 5(c)(vervolg / *continue*) / ...

(c) Beskou die volgende opstelling:
Consider the following setup:



Indien die statiese wrywingskoëffisiënt $\mu_s = 0{,}4$ is moet ons die waardes van m_2 bepaal wat die sisteem in rus sal hou.
If the coefficient of static friction is $\mu_s = 0{,}4$ we have to determine the range of values for m_2 to keep the system at rest.

(i) As m_2 klein is, sal m_1 neig om af te gly. / *With m_2 small, m_1 would tend to slide down.*

Die som van die kragte in y-riging op m_1 is : / *The sum of the forces in y-direction on m_1 is:*

____ (1)

Die som van die kragte in x-riging op m_1 is : / *The sum of the forces in x-direction on m_1 is:*

____ (1)

(ii) Spankrag T in die tou weens m_2: / *Tension in the rope due to m_2:*

____ (1)

(iii) Die normaalkrag N op m_1 is: / *The normal force N on m_1 is:*

____ (1)

(iv) Met $m_1 = 10$ kg, toon aan dat $m_2 = 2{,}8$ kg moet wees. / *With $m_1 = 10$ kg, show that $m_2 = 2{,}8$ kg.*

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____ (4)

(v) As m_2 groot is, sal m_1 neig om op te beweeg.
With m_2 large, m_1 *would tend to move upwards.*

Die som van die kragte in die x-rigting op m_1 (gee in simbole): / *The sum of t. forces in the x-direction on* m_1 *(give in symbols)*:

(4

(vi) Bepaal nou m_2 met $m_1 = 10$ kg / *Now, determine the value of* m_2 *with* $m_1 = 10$ kg

(4)

(vii) As $m_2 = m_1 = 10$ kg. en $\mu_k = 0,3$ is, toon aan die versnelling van die sisteem is $a = 0,78\ m/s^2$.
With $m_2 = m_1 = 10$ kg. and $\mu_k = 0,3$, *show that the acceleration of the system is* $a = 0,78\ m/s^2$.

(6)